

S&DS 265 / 565

Introductory Machine Learning

# Language Models

February 23

Yale

# Business items

- Assignment 3 is out
- Quiz 3 on Wednesday

# Midterm review sessions

Ruiqi: Friday, Feb 27, 7:00pm

Jianliang: Saturday, Feb 28, 5:00pm

Dingyao: Monday, March 2, 8:00pm

John: Tuesday, March 3, 5:30pm

Locations TBA

# Invitation: Panel discussion

Class on Monday, April 20:

We will have a panel discussion on

*Societal issues for AI and Machine Learning*

If you would like to volunteer to participate, please email us:

To: `sds265@yale.edu`

Subject: `iML panel`

# For Today

- Language models
- Concepts...no new methods

# Language models

- A language model is a way of *generating* any sequence of words

$$\begin{aligned} P(\text{"the whole forest had been anesthetized"}) = & \\ & P(\text{"the"}) \times P(\text{"whole"} \mid \text{"the"}) \\ & \times P(\text{"forest"} \mid \text{"the whole"}) \\ & \times P(\text{"had"} \mid \text{"the whole forest"}) \\ & \times P(\text{"been"} \mid \text{"the whole forest had"}) \\ & \times P(\text{"anesthetized"} \mid \text{"the whole forest had been"}) \end{aligned}$$

## Remixing Noon

Text generated from Channel Skin by Jeff Noon

"The whole forest had been anesthetised, her temples wired, her senses stimulated, her eyes were not on Eva, not on Eva, not on Eva, not on anybody in that same realm, the land of dreams and nightmares."

Viability: 0.000000326%



<https://chatbotslife.com/notes-on-remixing-noon-generative-text-and-markov-chains-84ff4ec23937>

# Text generation

- Words generated one-by-one
- A word is chosen by sampling from a probability distribution
- Result is purely synthetic text—generative AI



# Uses of language models

- Speech recognition
- Machine translation
- Text compression
- Texting
- Email completion
- Image captioning
- Mind reading from fMRI

# Dasher: LMs for assistive devices

Language models enable new modes of text input:

[https://www.youtube.com/watch?v=quw\\_Kci4fUg](https://www.youtube.com/watch?v=quw_Kci4fUg)

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- What are some ways of reducing the number of parameters?

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- Number of parameters:  $O(V \cdot \text{number of groups})$
- What are some example groupings?

# Grouping histories

- Unigrams:  $g(w_1, \dots, w_n) = \emptyset$ .

---

The notation  $O(\cdot)$  is called “Big Oh” and means “no greater than a constant times”, so that  $O(f(n))$  means a sequence that is bounded by  $C \cdot f(n)$  for large enough  $n$ .

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- Number of parameters grows as  $O(V)$ ,  $O(V^2)$ , and  $O(V^3)$ , respectively.

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# Estimating parameters

- The maximum likelihood estimate of a trigram model:

$$\hat{p}(w_3 | w_1, w_2) = \frac{\text{count}(w_1, w_2, w_3)}{\text{count}(w_1, w_2)}$$

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- $\text{count}(w_1, w_2, w_3)$  is just the number of times the trigram appears in a large corpus of text.



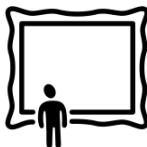
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- $\text{count}(w_1, w_2, w_3)$  is just the number of times the trigram appears in a large corpus of text.
- What are some problems with this model?

# Sparse data problem



The next group of slides presents one way of quantifying the problem of sparse data in language modeling

# Half Earth

Aug 06, 2018 by Foundation Staff 0 Comment Google Earth, Half-Earth Project, Map of Life

By Jeremy Malczyk, Michelle Duong, Ajay Ranipeta, Chris Heltne, Walter Jetz of Map of Life, Yale University, and the E.O. Wilson Biodiversity Foundation Half-Earth Project

This article originally in *Medium*, July 30, 2018



Learn how you can be part of bringing Half-Earth to life.

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## The Significance of Biodiversity



Narrow-billed Tody, *Todus angustirostris*. Photo by Julie Hart.

<https://eowilsonfoundation.org/mapping-species-for-half-earth/>

<https://www.half-earthproject.org/>

# Specious species



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- What is the chance I'll find a new species?
- What if Wilson observes 100 unique species?

# Missing species: Good-Turing

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- Good-Turing: Estimate of the probability that the next animal is a new species?  $\hat{p}_{GT} = 100/100,000 = 10^{-3}$ .
- This is an estimate of the missing probability mass.

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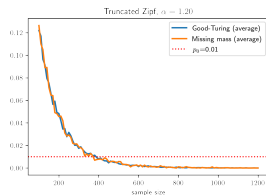
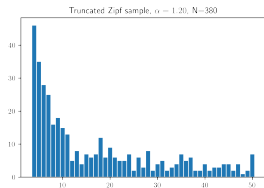
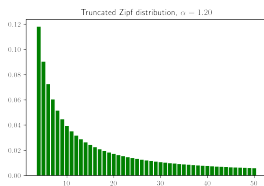
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- The MLE is supported on the observed data. We need to spread out the probability over unseen events.

# Good-Turing is remarkably accurate



Good-Turing estimate of the missing mass for a heavy-tailed distribution

# Estimating parameters

- The maximum likelihood estimate of a trigram model:

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- How can the model be strengthened?

# Smoothing and Interpolation

Smoothing:

$$\tilde{p}(w_3 | w_1, w_2) = \frac{\text{count}(w_1, w_2, w_3) + \varepsilon}{\text{count}(w_1, w_2) + V\varepsilon}$$

# Smoothing and Interpolation

Linear interpolation:

$$p(w_3 | w_1, w_2) = \lambda_3 \hat{p}(w_3 | w_1, w_2) + \lambda_2 \hat{p}(w_3 | w_2) + \lambda_1 \hat{p}(w_3)$$

where  $\lambda_1 + \lambda_2 + \lambda_3 = 1$ .

This is a type of “mixture model”

# How good is a language model?

The next group of slides presents a useful way of quantifying how good a language model is

## Recall: Geometric mean

The *arithmetic mean* of  $1/4$ ,  $4$ ,  $8$  is

$$\frac{1}{3} \left( \frac{1}{4} + 4 + 8 \right) = 4.0\overline{83}$$

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The geometric mean is no greater than the arithmetic mean

## Recall: Geometric mean

The *geometric mean* of  $x_1, \dots, x_n$  is

$$\sqrt[n]{x_1 x_2 \cdots x_n} = \left( \prod_{i=1}^n x_i \right)^{\frac{1}{n}}$$



# How good is a language model? Perplexity

*Perplexity* is defined as

$$\begin{aligned}\text{Perplexity}(\theta) &= p_{\theta}(w_1 w_2 \cdots w_N)^{-\frac{1}{N}} \\ &= \left( \prod_{i=1}^N p_{\theta}(w_i \mid w_{1:i-1}) \right)^{-\frac{1}{N}}\end{aligned}$$

where  $w_1, w_2, \dots, w_N$  is a large chunk of text that wasn't used to train the language model.

# How good is a language model? Perplexity

- Perplexity is the inverse of the geometric mean of the word probabilities
- Intuitively, if the perplexity is 20, the actual next word is in the top 20 words of the model (on average).
- This is the (geometric) average “branching factor” for the model on real text

# perplexity

Ask anything...



Model ▾



💰 Finance

✈️ Travel

🔧 Troubleshoot

♥️ Health

🔍 Research

# Recall: Embeddings

We discussed embeddings as constructed from simple bigram language models:

$$\log \mathbb{P}_{\text{near}}(w_1, w_2) = \varphi(w_1)^T \varphi(w_2) + \text{constant}$$

where  $\varphi(w)$  is the *embedding vector* for word  $w$ .

# Learning embeddings

- This language model is trained using stochastic gradient descent.
- If we have  $d = 500$  dimensional embeddings, there are 500 trainable parameters for *each* word
- How are these numbers adjusted (trained)?

# Learning embeddings: SGD

Start with *random* embeddings. Then iterate many, many times:

- 1 Sample a pair of words  $w_1$  and  $w_2$  that appear nearby
- 2 Adjust the embedding parameters of each word:

$$\varphi(w_1) = \varphi(w_1) + \eta \varphi(w_2)$$

$$\varphi(w_2) = \varphi(w_2) + \eta \varphi(w_1)$$

The vectors are brought closer together in each step!

# Learning embeddings: SGD

Why is this? The *loss* is the *negative* log-probability:

$$-\log \mathbb{P}_{\text{near}}(\mathbf{w}_1, \mathbf{w}_2) = -\varphi(\mathbf{w}_1)^T \varphi(\mathbf{w}_2) + \text{constant}$$

The derivative of this loss with respect to  $\varphi(\mathbf{w}_1)$  is  $-\varphi(\mathbf{w}_2)$

SGD update adds a small step size  $\eta$  times *minus* the derivative:

$$\varphi(\mathbf{w}_1) = \varphi(\mathbf{w}_1) + \eta \varphi(\mathbf{w}_2)$$

# Modern language models

An LLM assigns a “score” to possible next words:

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The function  $s(v; w_{1:n})$  is learned on large amounts of text (unsupervised) using a type of deep neural network called a *transformer*.

# Modern language models

In fact, the score is computed as

$$s(v; \underbrace{w_1, \dots, w_n}_{\text{word history}}) = \varphi(v)^T g(w_1, \dots, w_n)$$

- the “grouping function”  $g(w_1, \dots, w_n)$  is like an embedding, that maps the word history to a high dimensional vector
- The vector  $\varphi(v)$  is an embedding of word  $v$

We'll see how this is done when we discuss neural networks

# Summary of today: Language models

- A language model is used to predict or generate the next word
- Used in many different applications
- Probabilities need to be “smoothed” to avoid zeros
- Perplexity is a measure of a language model’s predictive power